MILITARY SPECIFICATION SHEET

BOLT, STRAIGHT SHANK, EXPANDABLE ALUMINUM ALLOY SLEEVE, ALLOY STEEL PIN, RELIABILITY AND MAINTAINABILITY REQUIREMENTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bolt described herein shall consist of this specification sheet and the latest issue of MIL-B-8831.

1. The expandable sleeve bolt may be used in bolted connections that require a precision controlled clearance fit or controlled interference fit.

2. The bolt may be used in design, repair and modification. The bolt may be used for the replacement of MS21134, MS21250, NAS 624 through NAS 644 bolts and tapered or straight shank interference fit fatigue rated fasteners.

3. The bolt shall be modified only by the original bolt manufacturer.

4. The bolt shall be used with nuts in accordance with MIL-N-8922/1 and MIL-N-8985/3 and wrenching elements in accordance with MS33787.

5. For installation of protruding head bolts, preload shall be induced by "wrenching" the nut. The bolt head "wrenching element" is intended to be used to prevent rotation of the bolt shank.

6. The bolt shall not be used in joints that are routinely disassembled.

7. The bolt shall not be used if the pin and sleeve have been separated.

8. MIL-B-8831/2 and /3 bolts are intended for use in applications requiring high interference levels. They shall not be used where high interference residual tensile stresses may cause stress corrosion cracking, especially in the 7000 series aluminum tempers.

9. MIL-B-8831/4 and /5 bolts are intended for use in fatigue critical areas where controlled low interference levels are required due to material susceptibility to stress corrosion cracking.

FSC 5306

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-B-83171 Bolt Straight Shank, Expandable Aluminum Alloy Sleeve Alloy Steel Pin Reliability and Maintainability Reqmnts for

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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